

Bhushan Karande

Jersey City, NJ | (201) 268-9389 | [linkedin.com/in/bhushan-karande](https://www.linkedin.com/in/bhushan-karande) | bvkarande1995@gmail.com

TECHNICAL SKILLS

Programming Languages: Python, R, Java, Scala, C++, C, Bash, Javascript, HTML, CSS, SQL, Prompt Engineering

Data Science & Machine Learning Libraries: Pandas, NumPy, PyTorch, TensorFlow, Keras, JAX, PyTorch Lightning, Hugging Face, Scikit-learn, NLTK, Spacy, PySpark, A/B Testing, Data Mining

Models: GPT Series, LLama Series, BERT, Sentence Transformers, CNN, RNN, LSTM

Tools & Platforms: MS Excel, $\LaTeX$, Visual Studio, JIRA, Docker, Terraform, AWS, Azure, GCP, Flask, FastAPI, Streamlit, React, Linux, MySQL, VectorDBs: LlamaIndex; Docker, Kubernetes, MLflow, AirFlow, Jenkins, Git, Gitlab CI/CD

PROFESSIONAL EXPERIENCE

Software Engineer — ESI Group (Bangalore, India)

Aug 2019 – Dec 2021

- Contributed to development and maintenance of the core MVC framework, enhancing product stability and expanding functionality across multiple releases.
- Designed and implemented C# class-wizard features, reducing development time by approximately 500 man-hours and accelerating team delivery by 35%.
- Developed Python scripts to identify and resolve technical debt, cutting open debt tickets by 30% and improving code coverage by 15%.
- Engineered C++ APIs to optimize UI Explorer features, reducing average load time by 40% and boosting user interaction speed by 25%.

RESEARCH THESIS

Source Localization of Acoustic Emission Generated during Damage of Concrete Using Neural Networks

May 2019

- Conducted initial modeling by implementing a linear regression baseline and an SVM regression model identifying key feature correlations and error patterns.
- Transitioned to a Keras-based convolutional neural network (CNN) leveraging time-frequency representations, improving localization precision to 0.8 cm on test specimens (46 % reduction in error compared to SVM).
- Engineered a data pipeline in NumPy/Pandas for raw waveform normalization, spectrogram feature extraction, and stratified train/validation/test splits, reducing preprocessing time by 50 % and ensuring balanced representation for over 10,000 emission events.
- Designed performance dashboards in MS Excel and matplotlib to compare regression, SVM, and CNN approaches; applied stratified K-fold cross-validation (5 folds) and hyperparameter tuning, resulting in a 12 % increase in model robustness under varied noise conditions.

OPEN-SOURCE PROJECTS

University AI Assistant Chatbot

- Architected and released an open-source chatbot combining GPT-4o, Llama-3, and Claude backends. Leveraged LangChain orchestration, Ollama model hosting, and a Streamlit frontend to deliver an intuitive, extensible interface.
- Implemented Elasticsearch-powered RAG pipelines for dynamic document retrieval, enabling the bot to deliver up-to-date, context-aware responses and responsibly source information for university students, faculty, and applicants.

EDUCATION

Stevens Institute of Technology, New Jersey

Master of Science in Data Science

Relevant Courses: Applied Machine Learning, Numerical Linear Algebra, Statistical Methods, Natural Language Processing, Big Data Technologies, Optimization for Data Science, Deep Learning

Indian Institute of Science, Bangalore

Master of Technology in Civil Engineering (Structural)

Relevant Courses: Mathematics for Engineers, Optimization Methods, Uncertainty Modeling and Analysis

Sardar Patel College of Engineering, Mumbai, India

Bachelor of Technology in Civil Engineering

LEADERSHIP & ACHIEVEMENTS

- Served as Placement Coordinator for the Civil Department at IISc during the 2018–2019 academic year, coordinating campus placements and industry outreach.
- Secured National Rank 44 in the Graduate Aptitude Test in Engineering (GATE) for Civil Engineering in 2017.